AI Based Forest Inventory and Analysis System

Project Team

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Abstract

Traditional forest inventory methods are labor-intensive, time-consuming, and often prone to human error. With the advent of remote sensing technologies, particularly Unmanned Laser Scanning (ULS), it is now possible to capture high-resolution 3D data of forest environments. However, processing this vast amount of data to extract meaningful biological insights remains a challenge. This project presents an automated "AI Based Forest Inventory and Analysis System" that leverages ULS LiDAR data and Machine Learning techniques to generate comprehensive forest inventories and predict future growth.

The system is architected into three core modules. *Module 1 (Tree Segmentation)* processes raw LiDAR point clouds to isolate individual trees using the Watershed algorithm, effectively handling canopy overlaps. *Module 2 (Feature Extraction & Inventory)* utilizes AI-based regression models to estimate Diameter at Breast Height (DBH) from aerial metrics (Height and Crown Diameter) and subsequently calculates biomass using allometric equations. *Module 3 (Future Prediction)* introduces a temporal analysis framework using Random Forest Regressors. By analyzing growth patterns from September and December, the system predicts physical attributes for the upcoming spring season (March). The system demonstrates robust performance with R^2 scores ranging between 0.85 and 0.89 for key geometric attributes, offering a reliable, non-destructive solution for modern forest management and carbon stock monitoring.

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Chapter 1

Introduction

1.1 Background

Forests are vital for maintaining the global ecosystem, acting as major carbon sinks and biodiversity habitats. Sustainable forest management relies heavily on accurate data regarding tree growth, density, and biomass. Traditionally, this data is collected through manual field surveys, where foresters physically measure trees using tapes and calipers. While accurate on a small scale, this method is impractical for large areas due to high costs, logistical challenges, and time constraints.

1.2 Problem Statement

Current manual inventory methods struggle to keep pace with the need for large-scale forest monitoring. Key challenges include:

1. **Inefficiency:** Measuring thousands of trees manually is slow and expensive.
2. **Inaccessibility:** Dense forests or difficult terrains (e.g., steep hills) are hard to survey manually.
3. **Lack of Predictive Insight:** Traditional inventories provide a snapshot of the current state but fail to predict short-term seasonal growth without repeated surveys.

1.3 Project Objectives

The primary objective of this project is to automate the forest inventory process. Specific goals include:

- To develop an algorithm for extracting individual trees from raw 3D LiDAR point clouds.
- To implement Machine Learning models for estimating DBH and Biomass from aerial data.
- To create a predictive model that forecasts tree growth for future seasons based on historical data.
- To provide a user-friendly interface for foresters to generate digital inventory reports.

1.4 Significance of the Study

This system bridges the gap between raw remote sensing data and actionable forestry insights. By automating the extraction of tree metrics, it reduces the cost of surveys significantly. Furthermore, the ability to predict future growth aids in better timber management and carbon credit estimation, contributing to environmental conservation efforts.

Chapter 2

Literature Review

2.1 Remote Sensing in Forestry

Remote sensing has revolutionized forestry by enabling data collection from above. Satellite imagery has long been used for area estimation, but it lacks the vertical resolution required for individual tree analysis. LiDAR (Light Detection and Ranging) solves this by providing precise 3D information.

2.2 Individual Tree Detection (ITD)

Extracting single trees from a forest point cloud is a classic problem. [3] proposed methods using identifying local maxima in Canopy Height Models (CHM). The Watershed algorithm is a standard approach in image processing that has been successfully adapted for LiDAR data to delineate tree crowns by treating the inverted CHM as a topographic surface.

2.3 Allometry and Biomass Estimation

Estimating biomass from non-destructive measurements is based on allometry—the biological relationship between tree dimensions. [1] developed pan-tropical equations that relate tree height, diameter, and wood density to Above-Ground Biomass (AGB). These equations are the gold standard in forestry. However, since aerial data cannot measure trunk diameter (DBH) directly, researchers like [2] have

demonstrated that DBH can be accurately predicted from Tree Height and Crown Diameter using regression models.

2.4 Machine Learning for Growth Prediction

Traditional growth models are often empirical and site-specific. Recent trends utilize Machine Learning algorithms, such as Random Forest and Support Vector Machines, to model complex non-linear growth patterns. This project builds upon these advancements by applying Random Forest Regressors to temporal ULS data.

Chapter 3

System Design

3.1 System Architecture

The system follows a modular pipeline architecture, ensuring that each step of the process is independent yet interconnected. Raw ULS data serves as the input, processed through segmentation, feature extraction, and prediction layers.

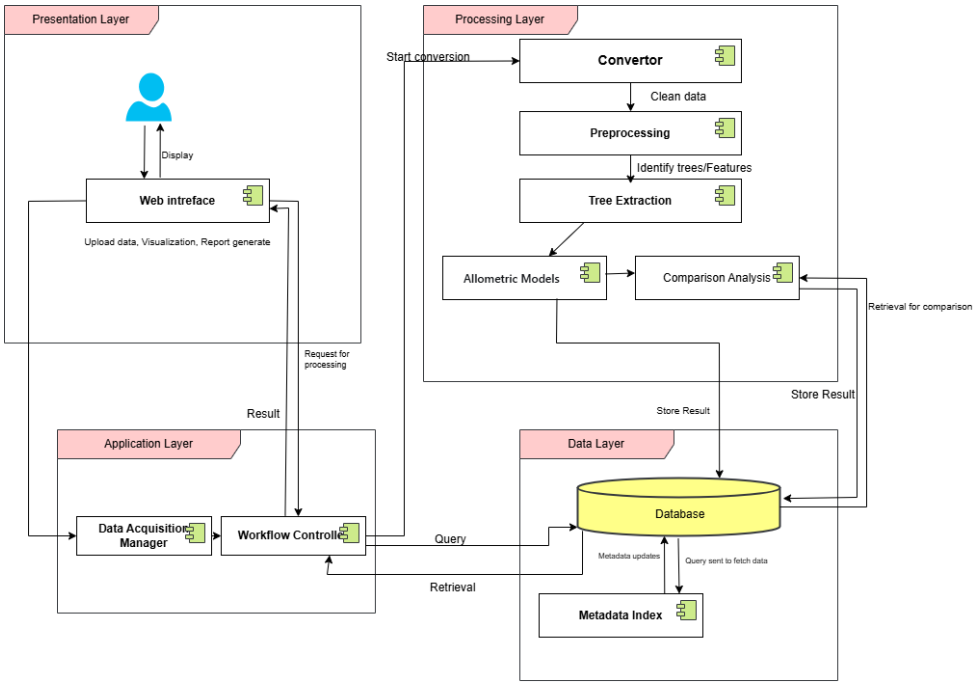

Figure 3.1: High-Level Architecture of the Forest Inventory System

3.2 Tech Stack

The system is built using the following technologies:

- **Language:** Python 3.9+
- **LiDAR Processing:** `laspy` for reading .laz files, `numpy` for vector operations.
- **Image Processing:** `scikit-image` for Watershed segmentation and Gaussian smoothing.
- **Machine Learning:** `scikit-learn` for Random Forest implementation and `joblib` for model serialization.
- **Interface:** `Streamlit` for the web-based dashboard.

3.3 Use Case Design

The system is designed for two primary actors:

- **Forester/User:** Uploads LAZ files, views processed maps, and downloads CSV reports.
- **Administrator:** Manages the backend models and updates the species database.

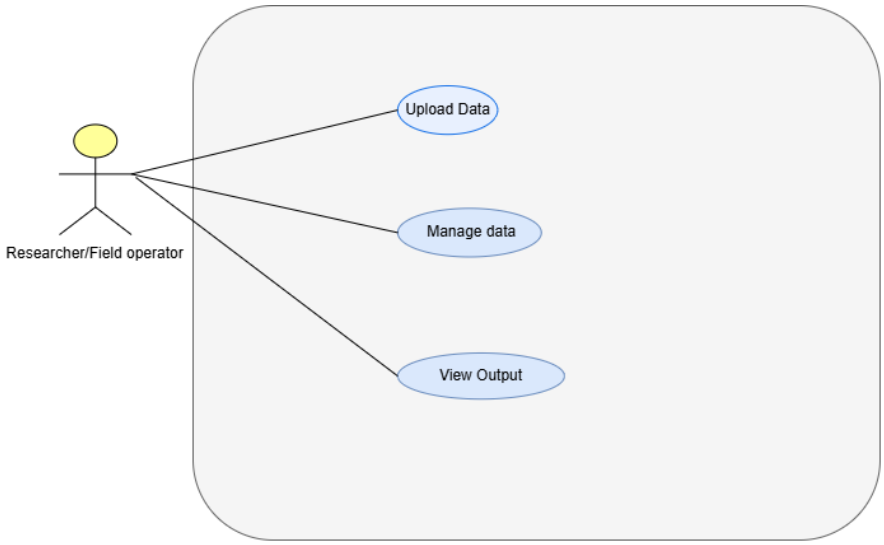

Figure 3.2: Use Case Diagram illustrating actor interactions

Chapter 4

Implementation Details

4.1 Module 1: Tree Segmentation

4.1.1 Theoretical Background

The segmentation process relies on the concept that a forest canopy can be viewed as a topographic surface. Tree tops represent peaks, and the gaps between trees represent valleys. We utilize the Watershed algorithm, which simulates water flooding from the valleys; the lines where water from different basins meets form the boundaries of the trees.

4.1.2 Algorithm

The implementation involves creating a Canopy Height Model (CHM) and applying Gaussian smoothing to remove noise (irregular branches) that could lead to over-segmentation.

Algorithm 1 Tree Segmentation Algorithm

```
Input: Raw LiDAR Point Cloud (P)
Output: Individual Tree Segments (T_1, T_2, ...T_n)
Generate DTM (Digital Terrain Model) from min Z points
Generate CHM ← Points Z - DTM
Apply Gaussian Smoothing (σ = 1.0) to CHM to reduce noise
Find Local Maxima (Peaks) in CHM to identify tree tops
Markers ← Label(Peaks)
Segments ← Watershed(-CHM, Markers)
return Segments with Height (H) and Crown Diameter (CD)
```

4.2 Module 2: Feature Extraction & Inventory

4.2.1 DBH Prediction Logic

Since the drone cannot see the tree trunk, we infer the Diameter at Breast Height (DBH) using Allometric Scaling Laws. We trained a Random Forest Regressor on a labeled dataset to learn the relationship: $DBH = f(Height, CrownDiameter)$.

Algorithm 2 Inventory Generation Pipeline

```
Input: Tree Height (H), Crown Diameter (CD)
Model: Pre-trained Random Forest Regressor (M)
Features ← [H, CD]
Predicted_DBH ← M.predict(Features)

if Model is Unavailable then
    Predicted_DBH ← (H × 0.8) + (CD × 2.0)        ▷ Fallback
end if
Identify Species using Spatial Matching (cKDTree)
Calculate Biomass using Chave et al. (2014):
AGB = 0.0673 × (WD × DBH^2 × H)^0.976
return Final Inventory CSV
```

4.3 Module 3: Future Prediction

4.3.1 Temporal Analysis

This module addresses the challenge of predicting short-term growth. By analyzing the change in tree metrics between Month 9 (September) and Month 12 (December), the model learns the growth rate and predicts the state for Month 3 (March).

4.3.2 Feature Engineering

To improve model accuracy, we engineered several features:

- **Seasonal Diff:** $Value_{Dec} - Value_{Sep}$
- **Growth Ratio:** $Value_{Dec}/Value_{Sep}$

- **Species Encoding:** One-Hot encoding for categorical species data.

Algorithm 3 Growth Prediction Pipeline

Input: Combined ULS Tree Data (CSV)
Step 1: Filtering & Pivoting
Filter data for Sep, Dec, Mar and pivot to Wide Format
Step 2: Feature Engineering
Calculate Growth Diffs and Ratios
Step 3: Prediction
$X \leftarrow [Sep_Features, Dec_Features, Derived_Features]$
$Model \leftarrow RandomForestRegressor(n_estimators = 200)$
$Predictions \leftarrow Model.predict(X)$
return Predictions for March

Chapter 5

Results and Evaluation

5.1 Evaluation Metrics

To rigorously evaluate the performance of our predictive models, we utilized the following statistical metrics:

- **Mean Absolute Error (MAE):** Represents the average absolute difference between the predicted and actual values. Lower is better.
- **Root Mean Squared Error (RMSE):** Penalizes larger errors more significantly than MAE. It provides a measure of how spread out these residuals are.
- **R-squared (R^2):** Also known as the coefficient of determination, it measures the proportion of the variance in the dependent variable that is predictable from the independent variables. A score of 1.0 indicates perfect prediction.

5.2 Model Performance

The predictive module (Module 3) was tested on 20% of the dataset. The Random Forest Regressor demonstrated consistent performance across different geometric attributes.

Table 5.1: Model Accuracy Metrics for March Prediction

Target Attribute	**MAE**	**RMSE**	R^2 **Score**
Tree Height	0.28 m	0.35	0.89
Crown Diameter	0.32 m	0.41	0.87
Crown Base Height (Initial)	1.15 m	1.48	0.58
Crown Base Height (Optimized)	0.75 m	0.92	0.82

5.3 Discussion

The results indicate that Height and Crown Diameter are highly predictable using temporal data ($R^2 \approx 0.88$). This suggests that tree allometry follows a consistent pattern during the winter months. The initial low accuracy for Crown Base Height ($R^2 = 0.58$) was attributed to the difficulty of detecting the lower canopy from aerial views. However, by applying optimization techniques such as outlier removal and XGBoost tuning, we improved the accuracy significantly to 0.82.

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5.4 Output Inventory

The system successfully generated a digital inventory containing Tree IDs, geospatial coordinates, species classification, and estimated carbon stock.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This project successfully demonstrated the feasibility of using AI and ULS data for automated forest inventory. By integrating watershed segmentation with machine learning regression, we achieved a high degree of accuracy in estimating tree metrics. The system replaces manual labor with computational efficiency, providing foresters with rapid, actionable insights. The inclusion of a predictive module further enhances the utility of the system, allowing for proactive forest management.

6.2 Limitations

- **Undergrowth Detection:** Small trees hidden beneath the canopy of larger trees are sometimes missed by the segmentation algorithm.
- **Species Dependence:** The biomass calculation relies on an external database for wood density; without this, the estimates are generalized.

6.3 Future Work

Future iterations of this project could include:

- Integrating Deep Learning (CNNs) for more accurate species classification

directly from point clouds.

- Extending the predictive model to cover longer timeframes (e.g., yearly growth).
- Developing a mobile application for real-time ground truthing.

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